

LIFE733 Assignment 2

Assignment Part 1a – Produce transcript sequences

1. Output of first function read_fasta_file in assignment2_1.py

[illegible]

2. Output of second function filter_cds in assignment2_1.py

```
C:\Users\Nikita_Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\venv\Scripts\python.exe "C:\Users\Nikita_Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\testing1.py"
{'LOC_0s01g01010.1': [[3448, 3615, '+'], [4356, 4454, '+'], [5456, 5559, '+'], [7135, 7943, '+'], [8027, 8149, '+'], [8231, 8319, '+'], [8407, 8607, '+'], [9209, 9616, '+'], [10103, 10186, '+'], [10273, 10296, '+']], 'LOC_0s01g01010.2': [[3448, 3615, '+'], [4356, 4454, '+'], [5456, 5559, '+'], [7135, 7943, '+'], [8027, 8149, '+'], [8231, 8319, '+']]}

Process finished with exit code 0
```

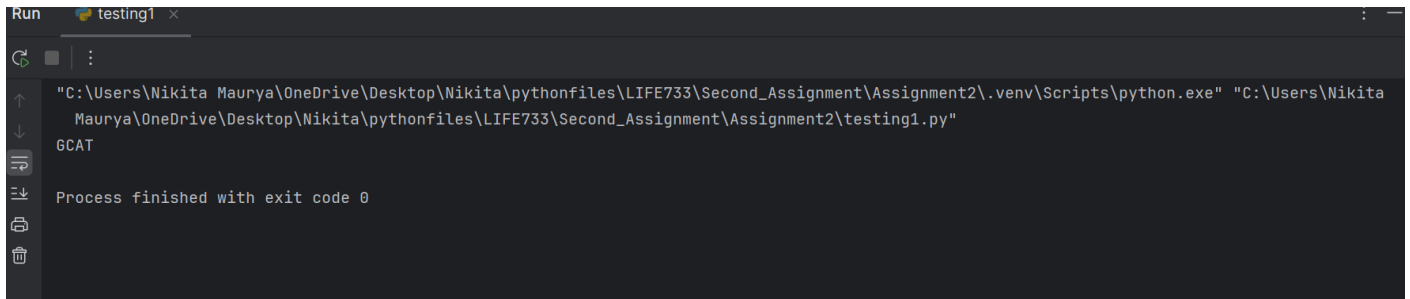
There are only two entries because made function to read only first 50 lines of gff.

Without stopping the counter at 50, this will be output

```
Run testing1 x
[43227443, 43227541, '+'], [43227625, 43227729, '+'], [43227831, 43227905, '+'], [43229647, 43229733, '+'], [43229912, 43229932, '+'],
'LOC_0s01g74640.2': [[43226327, 43226386, '+'], [43226613, 43226831, '+'], [43226948, 43227032, '+'], [43227224, 43227354, '+'], [43227443,
43227541, '+'], [43227625, 43227729, '+'], [43227831, 43227968, '+']], 'LOC_0s01g74640.3': [[43226709, 43226831, '+'], [43226948,
43227032, '+'], [43227224, 43227354, '+'], [43227443, 43227541, '+'], [43227625, 43227729, '+'], [43227831, 43227905, '+'], [43229647,
43229733, '+'], [43229912, 43229932, '+']], 'LOC_0s01g74650.1': [[43237673, 43237921, '-'], [43237161, 43237270, '-'], [43237022, 43237079,
'-'], [43235268, 43235585, '-'], [43235053, 43235190, '-'], [43234926, 43234977, '-'], [43234770, 43234849, '-'], [43233317, 43233376,
'-'], [43232631, 43232711, '-'], [43232359, 43232451, '-']], 'LOC_0s01g74650.2': [[43237673, 43237921, '-'], [43237161, 43237270, '-'],
[43237022, 43237079, '-'], [43235268, 43235531, '-'], [43235053, 43235193, '-'], [43234926, 43234977, '-'], [43234770, 43234849, '-'],
[43233317, 43233376, '-'], [43232631, 43232711, '-'], [43232359, 43232451, '-']], 'LOC_0s01g74650.3': [[43237673, 43237921, '-'],
[43237161, 43237270, '-'], [43237022, 43237079, '-'], [43235268, 43235531, '-'], [43235053, 43235190, '-'], [43234926, 43234977, '-'],
[43234770, 43234849, '-'], [43233317, 43233376, '-'], [43232631, 43232711, '-'], [43232359, 43232451, '-']], 'LOC_0s01g74660.1':
[[43244270, 43245253, '+'], 'LOC_0s01g74670.1': [[43250705, 43250709, '-'], [43250446, 43250601, '-'], [43248725, 43248837, '-'],
[43247347, 43247377, '-'], [43247205, 43247303, '-'], [43247064, 43247124, '-']], 'LOC_0s01g74680.1': [[43257619, 43259973, '+'],
[43260046, 43260876, '+'], [43260978, 43261529, '+'], [43261581, 43263296, '+']], 'LOC_0s01g74690.1': [[43267095, 43267664, '-'],
[43266403, 43267034, '-'], [43264405, 43266307, '-']]

Process finished with exit code 0
```

3. Output of third function reverse_fragment in assignment2_1.py

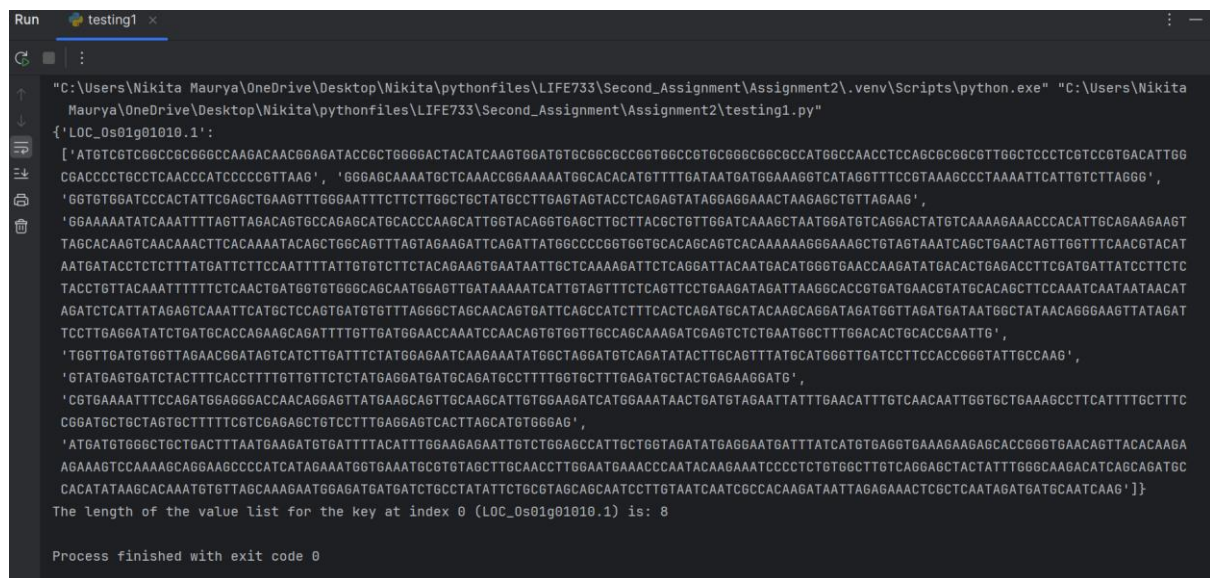


```
Run testing1 x
"C:\Users\Nikita Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\.venv\Scripts\python.exe" "C:\Users\Nikita Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\testing1.py"
GCAT
Process finished with exit code 0
```

4. Output of fourth function retrieve_fragment in assignment2_1.py

because made all my previous function to read only first 50 lines of .gff.

For testing I have only shown working with one locus id



```
Run testing1 x
"C:\Users\Nikita Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\.venv\Scripts\python.exe" "C:\Users\Nikita Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\testing1.py"
{"LOC_0s01g01010.1":
  ['ATGTCGTCGCGCCGCGGCAAGACACGAGATACCGCTGGGACTACATCAAGTGGATGTGCGCGCCGCGTGGCCGTCGCGGCGGCGCCATGGCCAACTCCAGCGCGCGCTTGGCTCCCTCGTCCGTGACATTGGCGACCCCTGCCCAACCATCCCCGTTAAG', 'GGGAGCAAAATGCTCAAACCGGAAAAATGGCACACATGTTTTGATAATGATGGAAGGTCATAGGTTTCCGTAAGGCCCTAAAATTCATTGCTCTTAAGG', 'GGTGGATCCCACTATTCGAGCTGAAGTTTGGGAATTTCTTCTTGGCTGCTATGCCCTTGAAGTAGTACCTCAGAGTATAGGAGGAACTAAGAGCTGTTAGAAAG', 'GGAAAAATATCAAAATTTAGTTAGACAGTGCAGAGCATGCAACCAAGCATTGGTACAGGTGAGCTTGCTTACGCTGTTGGATCAAGCTAATGGATGTCAGGACTATGTCAAAAGAAACCCACATTGCAGAAGAAGTTAGCAAGTCAACAACTTCACAAATACAGCTGGCAGTTTAGTAGAAGATTAGATTATGGCCCCGGTGGTGCACAGCAGTCACAAAAAGGAAAGCTGTAGTAATAGCTGAACATAGTTGGTTTCAACGTACATAATGATACCTCTCTTTATGATCTTCCAATTTTATTGTGCTCTACAGAAGTGAATAATTGCTCAAAGATTCTCAGGATTACAAATGACATGGGTGAACCAAGATATGACACTGAGACCTTCGATGATTATCCTTCTCTACCTGTTACAAATTTTCTCACTGATGGTGGGCAAGCAATGGAGTTGATAAAAAATCATTGTAGTTTCTCAGTTCTTGAAGATAGATTAAAGCACCCTGATGAACGTATGCACAGCTTCCAAATCAATAATAACATAGATCCTATTATAGATCAAAATTCATGCTCCAGTGAATGTTTAGGGCTAGCAACAGTATTCAAGCCATCTTCACTCAGATGCATACAGCAGGATAGATGGTTAGATGATAATGGCTATAACAGGGAAGTTATAGATTCCTTGAGGATATCTGATGCACCAAGCAGATTTTGTGATGGAACCAATCCAACAGTGTGGTTGCCAGCAAGATCGAGTCTCTGAAATGGCTTGGACACTGCACCGAATTG', 'GGTGTGATGGTTAGAACGGATAGTCACTTGTATTTCTATGGAGAATCAAGAAATATGGTGAAGTGCAGATATACTTGCAGTTTATGCATGGGTGATCCTTCCACCGGATTGCCAAG', 'GTATGAGTGAATCACTTTCACCTTTTGTGTTCTCTATGAGGATGATGCAAGTGCCTTTTGGTGTGTTGAGATGCTACTGAGAAAGGATG', 'CGTAAAAATTTCCAGATGGAGGACCAAGAGTTATGAAGCAATTGCAAGCATTGTGGAAGATCATGGAATAACTGATGTAGAAATTTTGAACATTTGTCAACAATTGGTGTGAAAGCCTTCATTTTGTCTTCGGATGCTGCTAGTGTCTTTTCTGTCGAGAGCTGTCTTTTGAAGGAGTCACTTATGATGTGGAG', 'ATGATGTGGGCTGCTGACTTTAATGAAGATGTGATTTTACATTTGGAAGAGAATTGTCTGAGCCATTGCTGGTAGATATGAGGAATGATTTATCATGTGAGGTGAAAGAGAGCACC6GTGAACAGTTACACAAGAAGAAAGTCCAAAGCAGGAAGCCCATCATAGAAATGGTGAATGCGTGTAGCTTGAACCTTGGATGAACCCCAATCAAGAAATCCCCTCTGTGGCTTGTCAAGAGCTACTATTTGGGCAAGACATCAGCAGATGCACATATAAGCACAATGTGTTAGCAAGAAATGGAGATGATGATCTGCTATATTCTGCTAGCAGCAATCCTTGTAAATCAATCGCCCAAGATAATTAGAGAACTCGCTCAATAGATGATGCAATCAAG']}
The length of the value list for the key at index 0 (LOC_0s01g01010.1) is: 8
Process finished with exit code 0
```

5. Output of fifth join_fragments in assignment2_1.py

because made all my previous function to read only first 50 lines of .gff

For testing I have only shown working with one locus id

[illegible]


```
Run testing1 x
"C:\Users\Nikita Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\.venv\Scripts\python.exe" "C:\Users\Nikita Maurya\OneDrive\Desktop\Nikita\pythonfiles\LIFE733\Second_Assignment\Assignment2\testing1.py"
LOC_Os01g01010.1:
MSSAAGQDN6DTAGDYIKWCMGAGGAGGAMANLQRGV6SLVRDIGDPCLNPSVKGSKMLKPEKWHOTCFDNDGKVI6FRKAKFIVL6GVDPITRAEVWEFL6GYALSSTSEYRKKLRAVRREKYQILVRQCSQMHP
SIGTGELAYAV6SKLMDVTRMTSKETHIAEEVSTSQTSQNTAGSLVEDSDYGP6GAQQSQKRESCSKSAELV6FNVHNDTSLYDSSNFIVSSTEVNCKSDSQDYNMGEPYDTETFDYPSLPVTNFFSTD6V6SNG
VDKNHCFSFVPEDRLRHRDRMHSFIQNNIDLIIESNCSDDVFRASNSDSAIFHSDAYKQDRWLDDNGYNREVIDSLRISDAPEADFV6GKSNVSVASKDRVSEWLTLHRIVDVVVRTDHLDFYGESRNMARMS
DILAVYAWVDPSTGYCQ6MSDLLSPFVVLVEDDADAFWCFEMLLRRMRENFMEGPTGVMKQLQALWKIMEITDVELFEHLSTIGAESLHFAFRMLLVLFRRRELSFEESLSMHEMMWAADFNEVDILHEENCLEPLLV
DMRNDLSCEVKEEHRVNSYTRRKSRSKRPHHRNGEMRVACNL6MKPNTRNPLCGLSGATIWARHQMQPHISTNVLAKNGDDLPFCVAAILVINRHKIIRETRSIDDAIK

Process finished with exit code 0
```

8. Output of assignment2_b.py

There are total 16 sequence out of 8023

```
outputfile_b.fasta
File Edit View
>LOC_Os01g11350.1 motif_pos = 201
MAQLPPKIPMTATANPEFGGGHHHAAHGHHRSPSMGAFLAAPLPFPPLPPAPANGAQQQQQQHQP5WVDFLDFSATKRGHRRSVSDSVAFLDPVSDNAGVGAHDFDRDDDDQLMSMFSDQLQPPPPQQQPAAPASASPSDHNSM
NDEKQDKGETDEAQSECDGATPGQPASPATVDPKRVKRIANRQSAQRSVRKLVQISELERSVSLQTEVSALSPRVAFLDHQRSLTLGNSHLKQRIAAALQDKIFKDGGETEGDREAAANLPPAKPQERGIPTGGRGPGPRPRQCDPRDRQGG
RRGRAMPALVIGRDPDAL*
>LOC_Os01g17260.1 motif_pos = 56
MADMSPRDTSDDTDNHLMEPGQALAAASDSDRSKDKHEDQKTLRLLAQNREAAKSRRLKAYVQLENSRLKLTQLEQLQRARQQGIFISSVQDTHSM5GNGALAFDMEYARWLEEHNRLINELSAVNAHAGDNLRGVVDKIMSHYE
EIFKQKGNAAKADVHVLSGMMKTPAERCFLWLGGRPSELLKLLSTQLEPLTEQQLSGIANLQSSQQAEDALSQGMALQQSLAETLAGSLGSSGSTGNVANYMGQMAMAMGKLTLENFLRQADNLRQQTLLQMQRILTTTQSRARLLVTSIDY
SSRLRALSSLLARLAPKE*
>LOC_Os01g17260.2 motif_pos = 56
MADMSPRDTSDDTDNHLMEPGQALAAASDSDRSKDKHEDQKTLRLLAQNREAAKSRRLKAYVQLENSRLKLTQLEQLQRARQQGIFISSVQDTHSM5GNGALAFDMEYARWLEEHNRLINELSAVNAHAGDNLRGVVDKIMSHYE
EIFKQKGNAAKADVHVLSGMMKTPAERCFLWLGGRPSELLKLLSTQLEPLTEQQLSGIANLQSSQQAEDALSQGMALQQSLAETLAGSLGSSGSTGNVANYMGQMAMAMGKLTLENFLRQADNLRQQTLLQMQRILTTTQSRARLLVTSIDY
SSRLRALSSLLARLAPKE*
>LOC_Os01g36220.1 motif_pos = 53
MASPGVCTTPTTSSAGSDQVAGGGAARRPVAPVAMEERKRKRKESNRLSAQRSRARKQQQLDELQAGVAAALRARNALGALAAAREARRCAAVRAENELLRARSVELAARLSDTLDAQLCHDAAAASSFAAPPMMAGAAATFPTGAVGAA
AGFMMPQLPPLLDATTMGHMCNYYY*
>LOC_Os01g46970.1 motif_pos = 238
MGNDIAVVTQKMGKAPSPKDPHALSPYLDWSNMQAYYGPILPPTFFSPGIAAGHTPPPFILGPQLVPSAFGKYAAIYPPGAFSHFPMPLMVSPLSMEPAKSVNSKSDSCSNKKMKEIDGAAVSTGSGNSEKTS6DCSLEGGSDGNQKASGT
PKKRSIDDRPKSGVETGGALTNDPRPSEQAALPNLCIPVTAIKPDVSTASDFRVIATPVTEVPTKDDKESKRERRKQSNRESARRSRLRQQAETEELARKVELLTAENTSLRREISRLTESSKRLRENSALMEKLTETGPDEAQEVPPVTKAQQ
ARGVENFLSMIDKTGTTPRSSGHMDHAIATPKLQQLGSLATDAVAAR*
>LOC_Os01g46970.2 motif_pos = 238
MGNDIAVVTQKMGKAPSPKDPHALSPYLDWSNMQAYYGPILPPTFFSPGIAAGHTPPPFILGPQLVPSAFGKYAAIYPPGAFSHFPMPLMVSPLSMEPAKSVNSKSDSCSNKKMKEIDGAAVSTGSGNSEKTS6DCSLEGGSDGNQKASGT
PKKRSIDDRPKSGVETGGALTNDPRPSEQAALPNLCIPVTAIKPDVSTASDFRVIATPVTEVPTKDDKESKRERRKQSNRESARRSRLRQQAETEELARKVELLTAENTSLRREISRLTESSKRLRENSALMEKLTETGPDEAQEVPPVTKAQQ
ARGVENFLSMIDKTGTTPRSSGHMDHAIATPKLQQLGSLATDAVAAR*
>LOC_Os01g55150.1 motif_pos = 150
MAQLPPKIPVAAPGHQHMASAGGADAAHAEFAEFAASRRGAHRRSLSDSVAFVEVAPAGCAGGEGFDRDDQLMSMFDPDEGGSSAPGSDNGGSDSDGGGDKHAAAGSDGQHAAGEPTQEQAATSPTELIRDPKRVKRIANRQSAQRSRV
RKLVQISELERSVITLQNEVSLSPRVAFLDQQRITLTVGNSHLKQRIAAALQDKIFKDAHQEALRKEIERLRQVYQQNTKLSGGLAADHAHVHGGPPPVRAEKELMS*
>LOC_Os01g59350.1 motif_pos = 198
MEGGRLGGAASASASAAAADARGGMSGFAAPQAHITNLNNVQPTQVDFGALAQSAGFRIEDLANLSTNGLFNLSNAHTIINDPLQFENYVKSISPSNITTTATVTVVDPQTLVPQGAQLNLVIRTGNVENMGESTIADTSPRTDSTDPDT
DERNQMFEGQQLAAPTASDSSDRSKDKLDHKTLLRLLAQNREAAKSRRLKAYIQNLESSRLKLTQIEQLQRARQQGIFISTSSDQSHSASGNGALAFDMEYARWLEEHNKINELRAAVNAHAGDNDLKSTVDSIMAHYNEIFKLKGVAAKADV
FHVLSGMMKTPAERCFLWLGGRFSELLKLLAQGLEPLTEQQLAGIANLQSSQQAEDALSQGMALQQSLAETLAGSLGAGSSGNVANYMGQMAMAMGKLTLENFLRQADNLRQQTLLQMQRILTTTQSRARLLAISDYFSRLRALSSLLWLA
RPRE*
>LOC_Os01g59350.2 motif_pos = 198
MEGGRLGGAASASASAAAADARGGMSGFAAPQAHITNLNNVQPTQVDFGALAQSAGFRIEDLANLSTNGLFNLSNAHTIINDPLQFENYVKSISPSNITTTATVTVVDPQTLVPQGAQLNLVIRTGNVENMGESTIADTSPRTDSTDPDT
DERNQMFEGQQLAAPTASDSSDRSKDKLDHKTLLRLLAQNREAAKSRRLKAYIQNLESSRLKLTQIEQLQRARQQGIFISTSSDQSHSASGNGALAFDMEYARWLEEHNKINELRAAVNAHAGDNDLKSTVDSIMAHYNEIFKLKGVAAKADV
FHVLSGMMKTPAERCFLWLGGRFSELLKLVRIYIIYIPTYFEASPGGKMLT*
>LOC_Os01g59760.1 motif_pos = 276
MASKAGGGGVARRRGGGRMRSLGRQGSMSYSLTLEVQSQLGEPLHSMNLDLRSVFPDGLAIADGAGATSSQQHQP6SGSLLRQGSITMPPELSKKTVDENVKIGIAAPKRMAETGGGGGGGRRRRRQPTLGEVTLDFLVKAGVVTQ6SLKELS
DVGNVDVPGRGVATGTGVDLPAGSHWIEQYKQQTASTDAHGGQGVQGYFNNRLVPQPLNVGPGALTEPSYSDGQTS5GMITGGMSDSQTPGRKRGMSGDVADKLMMERRQKRMKNRESAARSARKQAYTNELNENKVSRLREENVRLKQKLELD
RNRNENKVSRLREENVRLKQKLELD
Ln 1, Col 1 6,222 characters 100% Windows (CRLF) UTF-8
```